USB-C

J1
USB-C (USB4105)
VBUS
CC1
CC2
D−
D−
D+
D+
SHIELD
GND
SBU1
SBU2
A9
B9
A5
B5
A7
B7
A6
B6
A8
B8
USB_DM
USB_DP
R14 5.1k
R13 5.1k
SH
D5
TPD2S017
CH1_OUT
GND
CH1_IN
CH2_OUT
VCC
CH2_IN
USB_DM_ESP
USB_DP_ESP
VBUS_F
SMF5.0A
F1
1A fast (466)
D4
SS14
VBUS_F

POWER

U2
SGM2212-3.3
VIN
VOUT
GND
C2 10uF
C4 10uF

ESP32-C6 MCU

ESP32-C6-WROOM-1-N8
EN/CHIP_PU
GPIO0/ADC1_CH0/XTAL_32K_P
GPIO1/ADC1_CH1/XTAL_32K_N
GPIO2/ADC1_CH2
GPIO3/ADC1_CH3
MTMS/GPIO4/ADC1_CH4
MTDI/GPIO5/ADC1_CH5
MTCK/GPIO6/ADC1_CH6 U1
MTDO/GPIO7
NC
3V3
U0TXD/GPIO16
U0RXD/GPIO17
GPIO13/USB_D+
GPIO12/USB_D−
GPIO8
GPIO9
GPIO10
GPIO11
GPIO15
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
ESP32-C6-WROOM-1
I2S_WS
OC3_OUT
OC2_OUT
I2S_MCLK
I2S_DOUT
USB_DP_ESP
USB_DM_ESP
BOOT
I2S_DIN
I2S_BCLK
I2C_SDA
I2C_SCL
GATE2_DRV
GATE1_DRV
GATE3_DRV
OC1_OUT
C3 10uF
C6 100nF
R12 10k

POWER LED

D6
PWR
R15 1k

RESET / BOOT

R10 10k
C5 1uF
SW2
EN
R11 10k
SW1
BOOT

RELAY DRIVERS

R6 100
Q1 2N7002
D1 1N4148W
K1 G6K-2F-Y 4.5V
R9 10k
R4 100
Q2 2N7002
D2 1N4148W
R16 2.2k
K2 G6K-2F-Y 4.5V
R7 10k
R5 100
Q3 2N7002
D3 1N4148W
K3 G6K-2F-Y 4.5V
R8 10k
IN_P4
GATE1_PRE
GATE2_DRV
GATE3_DRV

BELL SENSE

SW3 Polarity
SW4 Polarity
SW5 Polarity
D7 1N4148W
D8 1N4148W
D9 1N4148W
OK1 LTV-217 (PC817)
OK2 LTV-217 (PC817)
OK3 LTV-217 (PC817)
OC1_IP
OC2_IP
OC3_IP
OC1_OUT
OC2_OUT
OC3_OUT
R17 5.1k
R1 5.1k
R2 5.1k
R3 1k
WF26 (6-way screw)

AUDIO ES8311 + LINE XFMR

U3
ES8311
CCLK
MCLK
PVDD
DVDD
DGND
SCLK/DMIC_SCL
ASDOUT
LRCK
DSDIN
AGND
EP
CE
CDATA
MIC1P/DMIC_SDA
MIC1N
VMID
ADCVREF
DACVREF
OUTN
OUTP
AVDD
R19 10k
R18 10k
R20 10k
C7 100nF
C8 100nF
C9 100nF
C10 10uF
C11 10uF
C12 1uF
C13 1uF
C14 1uF
C15 1uF
C16 1uF
C17 1uF
T1
SM-LP-5001

Sheet: /
File: doorbell.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 1/1